

Linear and Logistic Regression

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- **Core Predictive Tool:** Regression analysis is the fundamental method for predictive modeling, where a mathematical curve is fitted to historical observations to estimate and forecast future values.
- **Linear Regression as the Baseline:** The most basic and interpretable form of regression fits a straight line to the data, defining the expected response as:

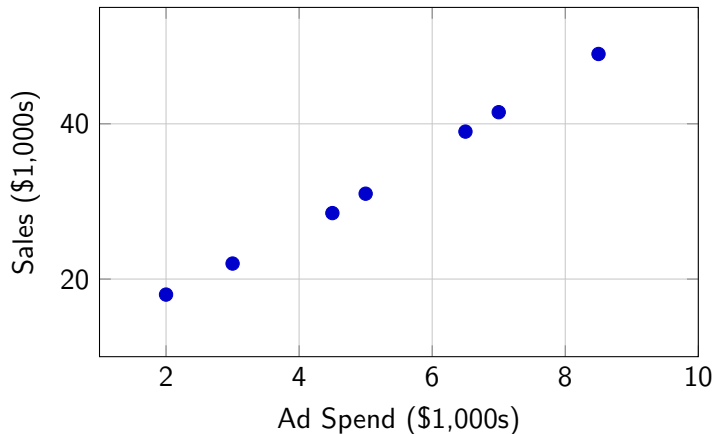
$$E[Y | x] = \alpha + \beta x$$

Real Life Instance

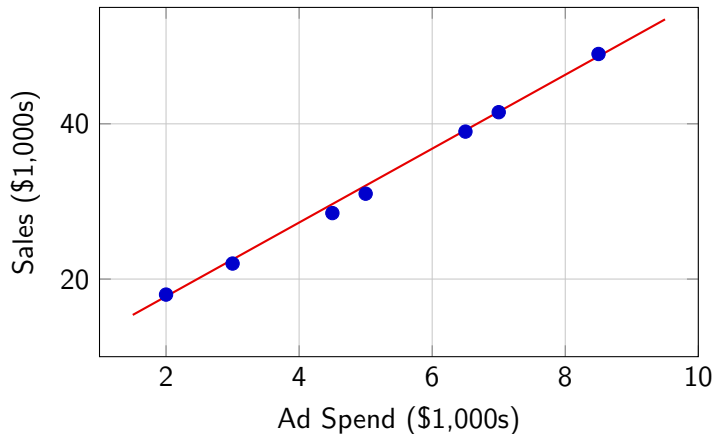
Context: Quarterly marketing spend (X , in \$1,000s) versus quarterly product revenue (Y , in \$1,000s) across 7 regional territories.

Region	Ad Spend (X)	Sales (Y)
A	2.0	18.0
B	3.0	22.0
C	4.5	28.5
D	5.0	31.0
E	6.5	39.0
F	7.0	41.5
G	8.5	49.0

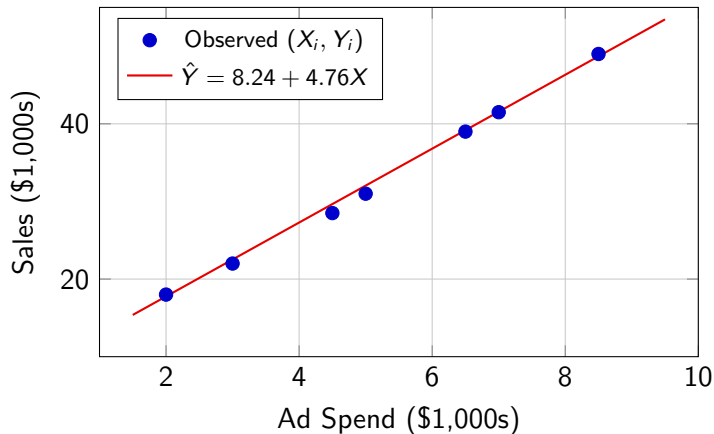
Plot and Fitting the Line



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Definition: Linear Regression

What is Linear Regression?

A statistical method to model the relationship between a dependent response variable Y and one or more explanatory variables (covariates) $X_1, X_2, \dots, X_p$:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p + \varepsilon$$

Crucial Distinction: Linearity in Parameters

- The model is strictly **linear in its parameters** ($\beta_0, \beta_1, \dots$), *not necessarily* in the covariates.
- Non-linear transformations of features (e.g., $Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \varepsilon$) are **still linear regression models** because the coefficients enter linearly.

Simple Linear Regression: Model & Assumptions

Model Formulation

For observations $i = 1, 2, \dots, n$, the simple linear regression model is expressed as:

$$Y_i = \alpha + \beta x_i + \varepsilon_i$$

where Y_i is the response variable, x_i is the fixed explanatory variable, and ε_i is the unobserved random error term.

Underlying Assumptions

Classical Underlying Assumptions (Gauss–Markov)

- **Zero Mean:** $\mathbb{E}[\varepsilon_i] = 0$ for all i , ensuring $\mathbb{E}[Y_i | x_i] = \alpha + \beta x_i$.
- **Homoscedasticity:** Constant error variance across all observations,

$$\text{Var}(\varepsilon_i) = \sigma^2 \quad \forall i.$$

- **Uncorrelated Errors:** $\text{Cov}(\varepsilon_i, \varepsilon_j) = 0$ for all $i \neq j$.
- **Fixed Regressors:** x_i are treated as known, non-random constants measured without error.
- **Normality (for inference):** $\varepsilon_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$, required for hypothesis tests (t -tests, F -tests) and confidence intervals.

OLS Derivation: Setup & Normal Equations

1. Objective Function

The Ordinary Least Squares (OLS) method chooses $\hat{\alpha}$ and $\hat{\beta}$ to minimize the Residual Sum of Squares (RSS):

$$S(\alpha, \beta) = \sum_{i=1}^n \varepsilon_i^2 = \sum_{i=1}^n (y_i - \alpha - \beta x_i)^2$$

2. First-Order Necessary Conditions

Setting the partial derivatives with respect to α and β equal to zero:

$$\frac{\partial S}{\partial \alpha} = -2 \sum_{i=1}^n (y_i - \alpha - \beta x_i) = 0 \quad (1)$$

$$\frac{\partial S}{\partial \beta} = -2 \sum_{i=1}^n x_i (y_i - \alpha - \beta x_i) = 0 \quad (2)$$

OLS Derivation: Closed-Form Estimators

Solving for Intercept ($\hat{\alpha}$)

Dividing the first normal equation (N1) by n :

$$\bar{y} = \hat{\alpha} + \hat{\beta}\bar{x} \implies \hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}$$

Key Insight: The fitted regression line always passes through the centroid $(\bar{x}, \bar{y})$.

Solving for Slope ($\hat{\beta}$)

Substitute $\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}$ into the second normal equation (N2):

$$\sum_{i=1}^n x_i y_i = (\bar{y} - \hat{\beta}\bar{x}) \sum_{i=1}^n x_i + \hat{\beta} \sum_{i=1}^n x_i^2$$

Rearranging terms and using $\sum_{i=1}^n x_i = n\bar{x}$:

$$\begin{aligned} \sum_{i=1}^n x_i y_i - n\bar{x}\bar{y} &= \hat{\beta} \left(\sum_{i=1}^n x_i^2 - n\bar{x}^2 \right) \\ \Rightarrow \hat{\beta} &= \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{S_{xy}}{S_{xx}} \end{aligned}$$

Regression of Y on X vs. X on Y : Intersection

Two Regression Models

- **Regression of Y on X :** Minimizes vertical squared errors

$$(y - \bar{y}) = b_{yx}(x - \bar{x}), \quad \text{where } b_{yx} = \frac{S_{xy}}{S_{xx}}$$

- **Regression of X on Y :** Minimizes horizontal squared errors

$$(x - \bar{x}) = b_{xy}(y - \bar{y}), \quad \text{where } b_{xy} = \frac{S_{xy}}{S_{yy}}$$

Regression of Y on X vs. X on Y : Intersection

Proof: Intersection at Centroid $(\bar{x}, \bar{y})$

Substitute the point $(x, y) = (\bar{x}, \bar{y})$ into both equations:

$$\text{Line 1 (Y on X): } \bar{y} - \bar{y} = b_{yx}(\bar{x} - \bar{x}) \implies 0 = 0 \quad \checkmark$$

$$\text{Line 2 (X on Y): } \bar{x} - \bar{x} = b_{xy}(\bar{y} - \bar{y}) \implies 0 = 0 \quad \checkmark$$

Since $(\bar{x}, \bar{y})$ satisfies both equations simultaneously, the two regression lines always intersect at the sample means $(\bar{x}, \bar{y})$.

Relationship Between Slopes and Correlation

Slopes of the Two Regression Lines

Expressed in terms of sample standard deviations s_x , s_y and Pearson's r :

$$b_{yx} = \frac{S_{xy}}{S_{xx}} = r \frac{s_y}{s_x}, \quad b_{xy} = \frac{S_{xy}}{S_{yy}} = r \frac{s_x}{s_y}$$

Proof: Product of Slopes

Multiplying the two estimated regression coefficients:

$$b_{yx} \cdot b_{xy} = \left(\frac{S_{xy}}{S_{xx}} \right) \left(\frac{S_{xy}}{S_{yy}} \right) = \frac{S_{xy}^2}{S_{xx} S_{yy}}$$

By definition of the sample correlation coefficient $r = \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}}$:

$$b_{yx} \cdot b_{xy} = r^2$$

Sampling Properties: Slope Estimator ($\hat{\beta}$)

Linear Representation

Define weights $w_i = \frac{x_i - \bar{x}}{S_{xx}}$ such that $\sum w_i = 0$, $\sum w_i x_i = 1$, and $\sum w_i^2 = \frac{1}{S_{xx}}$:

$$\hat{\beta} = \sum_{i=1}^n w_i Y_i$$

Expectation: Unbiasedness

Since $\mathbb{E}[Y_i] = \alpha + \beta x_i$:

$$\mathbb{E}[\hat{\beta}] = \sum_{i=1}^n w_i (\alpha + \beta x_i) = \alpha \underbrace{\sum_{i=1}^n w_i}_0 + \beta \underbrace{\sum_{i=1}^n w_i x_i}_1 = \beta$$

Sampling Properties: Slope Estimator ($\hat{\beta}$)

Variance

By independence of observations ($\text{Var}(Y_i) = \sigma^2$):

$$\text{Var}(\hat{\beta}) = \sum_{i=1}^n w_i^2 \text{Var}(Y_i) = \sigma^2 \sum_{i=1}^n w_i^2 = \frac{\sigma^2}{S_{xx}}$$

Sampling Properties: Intercept Estimator ($\hat{\alpha}$)

Expectation: Unbiasedness

From the relation $\hat{\alpha} = \bar{Y} - \hat{\beta}\bar{x}$, where $\mathbb{E}[\bar{Y}] = \alpha + \beta\bar{x}$ and $\mathbb{E}[\hat{\beta}] = \beta$:

$$\mathbb{E}[\hat{\alpha}] = \mathbb{E}[\bar{Y}] - \bar{x}\mathbb{E}[\hat{\beta}] = (\alpha + \beta\bar{x}) - \beta\bar{x} = \alpha$$

Proof: $\text{Cov}(\bar{Y}, \hat{\beta}) = 0$

Linear Representations

Recall that both $\bar{Y}$ and $\hat{\beta}$ can be written as linear combinations of Y_i :

$$\bar{Y} = \sum_{i=1}^n \frac{1}{n} Y_i, \quad \hat{\beta} = \sum_{j=1}^n w_j Y_j \quad \text{where } w_j = \frac{x_j - \bar{x}}{S_{xx}}$$

Bilinear Expansion of Covariance

Using the bilinearity of covariance and the uncorrelated errors assumption ($\text{Cov}(Y_i, Y_j) = \sigma^2$ if $i = j$, and 0 otherwise):

$$\begin{aligned} \text{Cov}(\bar{Y}, \hat{\beta}) &= \text{Cov} \left(\sum_{i=1}^n \frac{1}{n} Y_i, \sum_{j=1}^n w_j Y_j \right) \\ &= \sum_{i=1}^n \sum_{j=1}^n \frac{w_j}{n} \text{Cov}(Y_i, Y_j) = \sum_{i=1}^n \frac{w_i}{n} \text{Var}(Y_i) \end{aligned}$$

Conclusion

Since $\text{Var}(Y_i) = \sigma^2$ for all i :

$$\text{Cov}(\bar{Y}, \hat{\beta}) = \frac{\sigma^2}{n} \sum_{i=1}^n w_i = \frac{\sigma^2}{n} \sum_{i=1}^n \frac{x_i - \bar{x}}{S_{xx}} = \frac{\sigma^2}{n S_{xx}} \underbrace{\sum_{i=1}^n (x_i - \bar{x})}_0 = 0$$

Sampling Properties: Intercept Estimator ($\hat{\alpha}$)

Variance

Expressing $\hat{\alpha} = \sum_{i=1}^n \left(\frac{1}{n} - \bar{x}w_i\right) Y_i$, or using $\text{Cov}(\bar{Y}, \hat{\beta}) = 0$:

$$\begin{aligned}\text{Var}(\hat{\alpha}) &= \text{Var}(\bar{Y}) + \bar{x}^2 \text{Var}(\hat{\beta}) - 2\bar{x} \underbrace{\text{Cov}(\bar{Y}, \hat{\beta})}_0 \\ &= \frac{\sigma^2}{n} + \bar{x}^2 \left(\frac{\sigma^2}{S_{xx}}\right) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right) = \frac{\sigma^2 \sum x_i^2}{nS_{xx}}\end{aligned}$$

Non-linear Relationship: Log-Linear Regression

Exponential Growth with Multiplicative Errors

Suppose X grows linearly while Y grows exponentially according to:

$$Y_i = A e^{\beta x_i} \cdot \delta_i, \quad A > 0, \delta_i > 0$$

Although this relationship is inherently non-linear in variables, it is intrinsically linear in parameters via an analytical transformation.

Non-linear relationship (Cont.)

Linearizing Transformation

Taking natural logarithms of both sides yields:

$$\ln(Y_i) = \ln(A) + \beta x_i + \ln(\delta_i)$$

Defining transformed variables and parameters:

$$Y_i^* = \ln(Y_i), \quad \alpha = \ln(A), \quad \varepsilon_i = \ln(\delta_i)$$

we recover the standard Simple Linear Regression framework:

$$Y_i^* = \alpha + \beta x_i + \varepsilon_i, \quad \text{with } \mathbb{E}[\varepsilon_i] = 0, \text{ Var}(\varepsilon_i) = \sigma^2$$

Multiple Linear Regression (MLR): Model

Model Formulation

For n observations and p explanatory variables ($X_1, X_2, \dots, X_p$):

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip} + \varepsilon_i, \quad i = 1, 2, \dots, n$$

where:

- Y_i is the response variable for the i -th unit.
- β_0 is the intercept parameter.
- β_j is the partial regression coefficient for regressor X_j .
- ε_i is the unobserved random error term.

Partial Effects and Core Assumptions

- **Ceteris Paribus Interpretation:**

$$\beta_j = \frac{\partial \mathbb{E}[Y \mid \mathbf{x}]}{\partial x_j}$$

Each β_j represents the expected change in Y per unit increase in X_j , **holding all other covariates fixed**.

- **Classical Error Assumptions:**

$$\mathbb{E}[\varepsilon_i] = 0, \quad \text{Var}(\varepsilon_i) = \sigma^2, \quad \text{Cov}(\varepsilon_i, \varepsilon_k) = 0 \quad (i \neq k)$$

MLR: Matrix Formulation

System in Matrix Form: $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$

$$\begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix} = \begin{bmatrix} 1 & x_{11} & \dots & x_{1p} \\ 1 & x_{21} & \dots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \dots & x_{np} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}$$

- $\mathbf{Y} \in \mathbb{R}^{n \times 1}$, $\mathbf{X} \in \mathbb{R}^{n \times (p+1)}$, $\boldsymbol{\beta} \in \mathbb{R}^{(p+1) \times 1}$, $\boldsymbol{\varepsilon} \in \mathbb{R}^{n \times 1}$.
- Assume $\text{rank}(\mathbf{X}) = p + 1 \leq n$ (no exact multicollinearity).

Derivation: Residual Sum of Squares (RSS)

Quadratic Loss Function

Minimize the sum of squared residuals with respect to β :

$$\begin{aligned} S(\beta) &= \varepsilon^\top \varepsilon \\ &= (\mathbf{Y} - \mathbf{X}\beta)^\top (\mathbf{Y} - \mathbf{X}\beta) \\ &= \mathbf{Y}^\top \mathbf{Y} - (\mathbf{X}\beta)^\top \mathbf{Y} - \mathbf{Y}^\top (\mathbf{X}\beta) + (\mathbf{X}\beta)^\top (\mathbf{X}\beta) \\ &= \mathbf{Y}^\top \mathbf{Y} - 2\beta^\top \mathbf{X}^\top \mathbf{Y} + \beta^\top \mathbf{X}^\top \mathbf{X} \beta \end{aligned}$$

(using the identity that scalar $\mathbf{Y}^\top \mathbf{X}\beta = \beta^\top \mathbf{X}^\top \mathbf{Y}$).

Derivation: Normal Equations & Solution

First-Order Conditions and Estimator

Setting the gradient vector with respect to β to zero:

$$\frac{\partial S(\beta)}{\partial \beta} = -2\mathbf{X}^\top \mathbf{Y} + 2\mathbf{X}^\top \mathbf{X} \beta = \mathbf{0}$$

This yields the **Normal Equations**:

$$\mathbf{X}^\top \mathbf{X} \hat{\beta} = \mathbf{X}^\top \mathbf{Y}$$

Since $\text{rank}(\mathbf{X}) = p + 1$, $\mathbf{X}^\top \mathbf{X}$ is symmetric and strictly positive definite, giving the unique solution:

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$$

Properties of the OLS Estimator $\hat{\beta}$

Theoretical Properties of $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$

- **Global Minimizer:** By construction, $\hat{\beta}$ minimizes the residual sum of squares:

$$S(\hat{\beta}) = \|\mathbf{Y} - \mathbf{X}\hat{\beta}\|^2 = \min_{\beta} \|\mathbf{Y} - \mathbf{X}\beta\|^2$$

- **Unbiasedness:** Provided $\mathbb{E}[\varepsilon \mid \mathbf{X}] = \mathbf{0}$, the estimator is strictly unbiased:

$$\mathbb{E}[\hat{\beta}] = \beta$$

- **Variance-Covariance Matrix:** Under homoscedasticity and uncorrelated errors ($\text{Cov}(\varepsilon) = \sigma^2 \mathbf{I}$):

$$\text{Var}(\hat{\beta}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}$$

- **Gauss–Markov Optimality (BLUE):** $\hat{\beta}$ has the smallest variance among all linear unbiased estimators.

Normality Assumption:

Properties:

If $\varepsilon \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$, then:

$$\hat{\beta} \sim \mathcal{N}(\beta, \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1})$$

Under Normality, the least square estimator of the parameter is equal to its maximum likelihood estimator.

ANOVA: Partitioning of Variability

Sum of Squares Decomposition

Total variability around the sample mean $\bar{y}$ is partitioned into explained and unexplained components:

$$SS_{\text{TOT}} = SS_{\text{REG}} + SS_{\text{RES}} \quad (\text{or } SS_{\text{ERR}})$$

- $SS_{\text{TOT}} = \sum_{i=1}^n (y_i - \bar{y})^2$: Total variation in the response.
- $SS_{\text{REG}} = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$: Variation explained by the regression model.
- $SS_{\text{RES}} = \sum_{i=1}^n (y_i - \hat{y}_i)^2$: Residual (unexplained / error) variation.

Proof: $SS_{\text{TOT}} = SS_{\text{REG}} + SS_{\text{RES}}$

Algebraic Decomposition

Express the total deviation as the sum of explained and residual components:

$$y_i - \bar{y} = (\hat{y}_i - \bar{y}) + (y_i - \hat{y}_i)$$

Squaring both sides and summing over all $i = 1, 2, \dots, n$:

$$\begin{aligned}\sum_{i=1}^n (y_i - \bar{y})^2 &= \sum_{i=1}^n [(\hat{y}_i - \bar{y}) + (y_i - \hat{y}_i)]^2 \\&= \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 + \sum_{i=1}^n (y_i - \hat{y}_i)^2 + 2 \sum_{i=1}^n (\hat{y}_i - \bar{y})(y_i - \hat{y}_i) \\&= SS_{\text{REG}} + SS_{\text{RES}} + 2 \underbrace{\sum_{i=1}^n (\hat{y}_i - \bar{y})e_i}_{\text{Cross-product term}}\end{aligned}$$

where $e_i = y_i - \hat{y}_i$ is the i -th OLS residual.

Proof: Vanishing of the Cross-Product Term

Orthogonality of Residuals

Expanding the cross-product term:

$$\sum_{i=1}^n (\hat{y}_i - \bar{y}) e_i = \sum_{i=1}^n \hat{y}_i e_i - \bar{y} \sum_{i=1}^n e_i$$

Using the OLS first-order conditions (for models with an intercept β_0):

- $\sum_{i=1}^n e_i = 0$ (residuals sum to zero).
- $\sum_{i=1}^n \hat{y}_i e_i = \hat{\beta}^\top \mathbf{X}^\top \mathbf{e} = \hat{\beta}^\top (\mathbf{0}) = 0$ ($\mathbf{e} \perp \text{Col}(\mathbf{X})$).

$$\implies \sum_{i=1}^n (\hat{y}_i - \bar{y}) e_i = 0 - 0 = 0$$

Therefore, the Pythagorean decomposition holds strictly:

$$SS_{\text{TOT}} = SS_{\text{REG}} + SS_{\text{RES}}$$

ANOVA: The Global Hypothesis Test

Testing Overall Model Significance

To determine whether the k explanatory covariates collectively contribute useful linear information about Y :

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_k = 0$$

$$H_1 : \beta_j \neq 0 \quad \text{for at least one } j \in \{1, 2, \dots, k\}$$

- Under H_0 , none of the explanatory variables linearly explain Y (reduced model: $Y_i = \beta_0 + \varepsilon_i$).
- Rejection of H_0 implies that at least one regressor has a non-zero linear relationship with Y .

The ANOVA Table for Multiple Regression

Degrees of Freedom and Mean Squares

Source	df	Sum of Squares	Mean Square (MS)
Regression	k	SS_{REG}	$MS_{\text{REG}} = \frac{SS_{\text{REG}}}{k}$
Residual	$n - k - 1$	SS_{RES}	$MS_{\text{RES}} = \frac{SS_{\text{RES}}}{n - k - 1}$
Total	$n - 1$	SS_{TOT}	—

- Total degrees of freedom ($n - 1$) partitions as $k + (n - k - 1)$.
- In statistical software packages, SS_{REG} is often subdivided into the individual contribution of each explanatory variable (sequential or partial SS).

The Global F -Test Statistic

Test Statistic Construction

The test statistic is the ratio of the mean squares:

$$F = \frac{MS_{\text{REG}}}{MS_{\text{RES}}} = \frac{SS_{\text{REG}}/k}{SS_{\text{RES}}/(n-k-1)} \sim F_{k, n-k-1} \quad (\text{under } H_0)$$

- A large value of F indicates that the explained variability dominates residual noise.
- **Decision Rule:** Reject H_0 at significance level α if $F > F_{\alpha; k, n-k-1}$, or if the corresponding p -value $< \alpha$.

Definitions and Mathematical Formulation

- **Coefficient of Determination (R^2):**

$$R^2 = 1 - \frac{SS_{\text{RES}}}{SS_{\text{TOT}}}$$

Measures the proportion of total variation in Y explained by the regressors.

- **Adjusted R^2 ($\bar{R}^2$ or R^2_{adj}):**

$$R^2_{\text{adj}} = 1 - \frac{SS_{\text{RES}}/(n - k - 1)}{SS_{\text{TOT}}/(n - 1)} = 1 - (1 - R^2) \left(\frac{n - 1}{n - k - 1} \right)$$

Penalizes model complexity by scaling sums of squares by their respective degrees of freedom.

Extracting ANOVA Results in R

For a fitted model object `model <- lm(y ~ x1 + x2 + ... + xk, data = df)`:

- `summary(model)`: Reports the global F -statistic, degrees of freedom (k and $n - k - 1$), and the overall model p -value on the final summary line.
- `anova(model)`: Displays the sequential ANOVA decomposition table, breaking down sum of squares, mean squares, and F -tests for individual predictors.

Motivation: Binary Response Variables

The Classification Problem

In many real-world settings, the response variable Y is **categorical/binary**:

$$Y_i \in \{0, 1\} \quad (\text{e.g., Default vs. Non-Default, Disease vs. Healthy})$$

Here, the goal is to model the conditional probability
 $p(x) = P(Y = 1 \mid X = x)$.

Why Ordinary Least Squares (LPM) Fails

Fitting a Linear Probability Model $Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$ has severe flaws:

- **Unbounded Predictions:** The fitted line $\hat{p}(x) = \hat{\beta}_0 + \hat{\beta}_1 x$ will predict values outside $[0, 1]$ for extreme x , violating axioms of probability.
- **Inherent Heteroscedasticity:** Since $Y_i \sim \text{Bernoulli}(p_i)$,

$$\text{Var}(\varepsilon_i \mid x_i) = p_i(1 - p_i) = (\mathbf{x}_i^\top \boldsymbol{\beta})(1 - \mathbf{x}_i^\top \boldsymbol{\beta}) \neq \text{constant}$$

- **Non-Normal Errors:** Residuals are discrete and bimodal:
 $e_i \in \{1 - p_i, -p_i\}$.

The Logistic Regression Solution

Sigmoid / Logistic Link Function

To restrict the output strictly to $(0, 1)$, map the linear predictor $\eta = \beta_0 + \beta_1 x$ through the **standard logistic function**:

$$p(x) = P(Y = 1 \mid X = x) = \frac{e^{\beta_0 + \beta_1 x}}{1 + e^{\beta_0 + \beta_1 x}} = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

Odds and the Logit Transformation

- **Odds of Success:** The ratio of success to failure probability:

$$\text{Odds} = \frac{p(x)}{1 - p(x)} = e^{\beta_0 + \beta_1 x} \in (0, \infty)$$

- **Logit Transformation (Log-Odds):** Taking the natural log linearizes the relationship in parameters:

$$\text{logit}(p(x)) = \ln \left(\frac{p(x)}{1 - p(x)} \right) = \beta_0 + \beta_1 x \in (-\infty, \infty)$$

Parameter Estimation: Maximum Likelihood (MLE)

Bernoulli Likelihood Setup

Let $Y_i \in \{0, 1\}$ be independent binary responses with success probabilities:

$$p_i = P(Y_i = 1 \mid \mathbf{x}_i) = \frac{1}{1 + e^{-\mathbf{x}_i^\top \boldsymbol{\beta}}}$$

The probability mass function for a single observation is Bernoulli:

$$P(Y_i = y_i \mid \mathbf{x}_i) = p_i^{y_i} (1 - p_i)^{1-y_i}, \quad y_i \in \{0, 1\}$$

Assuming independence, the overall Likelihood function is:

$$L(\boldsymbol{\beta}) = \prod_{i=1}^n p_i^{y_i} (1 - p_i)^{1-y_i}$$

Derivation: The Log-Likelihood Function

Log-Likelihood $\ell(\beta)$

Taking the natural logarithm of $L(\beta)$:

$$\begin{aligned}\ell(\beta) &= \sum_{i=1}^n [y_i \ln(p_i) + (1 - y_i) \ln(1 - p_i)] \\ &= \sum_{i=1}^n \left[y_i \ln \left(\frac{p_i}{1 - p_i} \right) + \ln(1 - p_i) \right]\end{aligned}$$

Substitute the logit relation $\ln \left(\frac{p_i}{1 - p_i} \right) = \mathbf{x}_i^\top \beta$ and $1 - p_i = \frac{1}{1 + e^{\mathbf{x}_i^\top \beta}}$:

$$\ell(\beta) = \sum_{i=1}^n \left[y_i (\mathbf{x}_i^\top \beta) - \ln \left(1 + e^{\mathbf{x}_i^\top \beta} \right) \right]$$

Solving the Score Equations

Score Function (First-Order Condition)

Differentiating $\ell(\boldsymbol{\beta})$ with respect to parameter vector $\boldsymbol{\beta}$:

$$\nabla_{\boldsymbol{\beta}} \ell(\boldsymbol{\beta}) = \sum_{i=1}^n \left(y_i - \frac{e^{\mathbf{x}_i^\top \boldsymbol{\beta}}}{1 + e^{\mathbf{x}_i^\top \boldsymbol{\beta}}} \right) \mathbf{x}_i = \sum_{i=1}^n (y_i - p_i) \mathbf{x}_i = \mathbf{0}$$

In matrix form:

$$\mathbf{X}^\top (\mathbf{Y} - \mathbf{p}) = \mathbf{0}$$

Predicting Discrete Classes: Decision Boundary

From Probability to Class Label

Logistic regression outputs a continuous conditional probability:

$$\hat{p}(\mathbf{x}) = P(Y = 1 \mid \mathbf{x}) = \frac{1}{1 + e^{-\mathbf{x}^\top \hat{\beta}}} \in (0, 1)$$

To assign a binary label $\hat{Y} \in \{0, 1\}$, apply a **decision threshold** $c \in (0, 1)$:

$$\hat{Y} = \begin{cases} 1, & \text{if } \hat{p}(\mathbf{x}) \geq c \\ 0, & \text{if } \hat{p}(\mathbf{x}) < c \end{cases}$$

The Default Threshold ($c = 0.5$)

- Assigns the observation to the more likely outcome
- On the log-odds (logit) scale, this corresponds to:

$$\hat{p}(\mathbf{x}) \geq 0.5 \iff \ln \left(\frac{\hat{p}(\mathbf{x})}{1 - \hat{p}(\mathbf{x})} \right) \geq 0 \iff \mathbf{x}^\top \hat{\boldsymbol{\beta}} \geq 0$$

- The hyperplane $\mathbf{x}^\top \hat{\boldsymbol{\beta}} = 0$ forms the linear **decision boundary**.

Thank You!

Topics Covered: Simple Linear Regression → Multiple Linear Regression → ANOVA & R^2_{adj} → Logistic Regression & Classification